



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Data Manipulation With Pandas

Name: Deva Dharshini

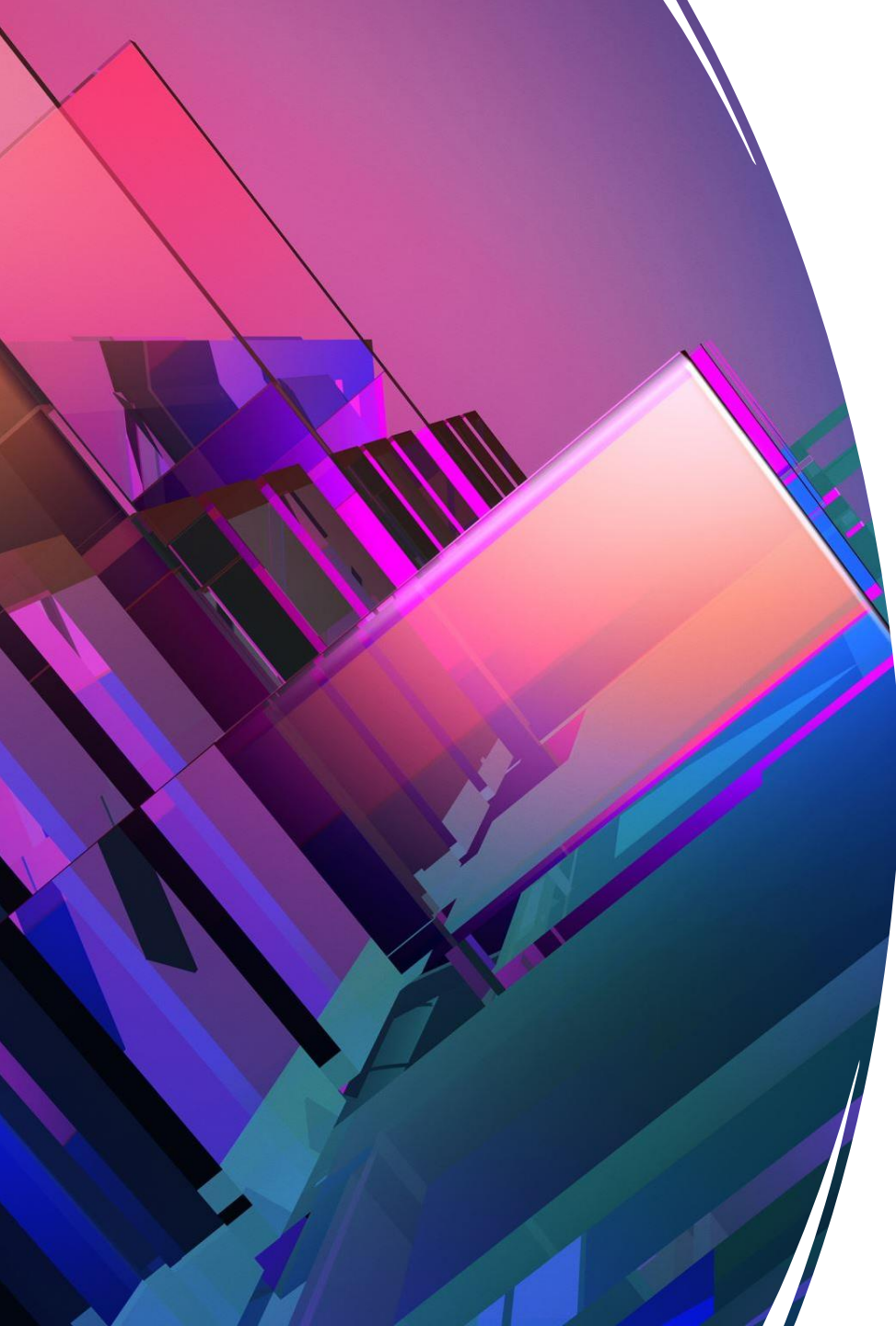
Register Number: 192472350

Course Code: DSA0404

Course Name: Fundamentals
of data science

Supervised by:

Dr. Krishnaraj M



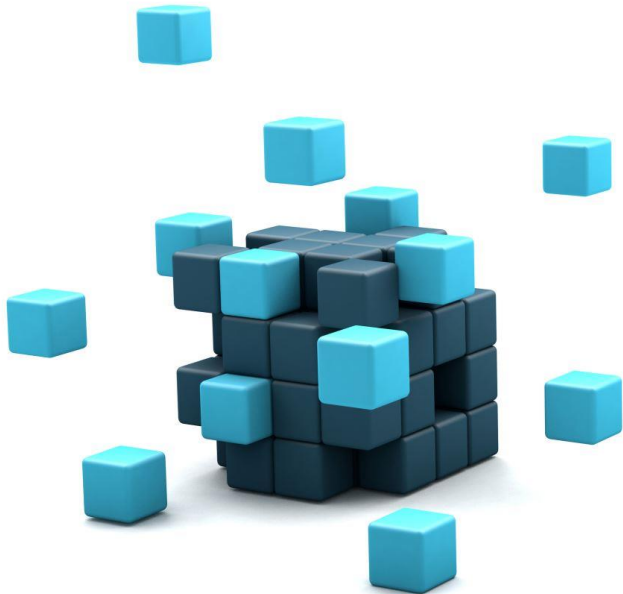
Introduction to Data Manipulation

- Data manipulation refers to the process of modifying and organizing raw data into a useful and structured form.
- Raw datasets often contain unnecessary information, missing values, and inconsistent records.
- Data manipulation helps select only the required information from a dataset.
- It allows data to be filtered, sorted, grouped, and transformed according to analysis requirements.
- Properly manipulated data makes further analysis and visualization easier.

Introduction to Pandas

- Pandas is an open-source Python library designed for working with structured and tabular data.
- It provides simple and efficient tools for reading, manipulating, and analyzing datasets.
- Pandas supports common file formats such as CSV, Excel, JSON, and SQL data.
- It provides powerful operations for selecting, filtering, sorting, grouping, and cleaning data.

Pandas Data Structures



Pandas mainly provides two important data structures for handling data.

Series

A Series is a one-dimensional structure that stores a sequence of values along with an index.

DataFrame

A DataFrame is a two-dimensional tabular structure consisting of rows and columns.

```
import pandas as pd
```

```
data = {  
    "Name":["Arun","Priya","Rahul"],  
    "Marks":[85,92,78]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

A DataFrame is commonly used to represent datasets in a table-like format.

Reading and Inspecting Data

Pandas provides functions to load datasets and understand their basic structure.

Reading a CSV Dataset

```
import pandas as pd  
df = pd.read_csv("students.csv")
```

Inspecting the Dataset

```
print(df.head())  
print(df.info())  
print(df.describe())
```

head() displays the first few records.

info() provides information about columns and data types.

describe() provides statistical information about numerical columns.

Selecting Data

Data selection allows specific rows and columns to be accessed from a DataFrame.

Selecting a Column

```
print(df["Name"])
```

Selecting Multiple Columns

```
print(df[["Name","Marks"]])
```

Selecting Rows

```
print(df.iloc[0:3])
```

Selection is useful when the analysis requires only particular attributes or records from a dataset.

Filtering Data



Filtering is used to extract records that satisfy a particular condition.



For example, students who scored more than 80 marks can be selected using:



```
high_marks = df[df["Marks"] > 80]
```



```
print(high_marks)
```



Filtering can also be performed using multiple conditions.



```
result = df[(df["Marks"] > 80) & (df["Age"] > 18)]
```



This operation helps focus the analysis on relevant records.



Handling Missing Data

- Missing data is common in real-world datasets and must be handled before analysis.
- **Detecting Missing Values**
- `print(df.isnull().sum())`
- **Removing Missing Records**
- `df = df.dropna()`
- **Replacing Missing Values**
- `df["Marks"] = df["Marks"].fillna(0)`
- Appropriate handling of missing values improves the quality and reliability of the dataset.

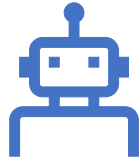
Sorting and Ranking Data

- Pandas allows records to be arranged according to the values of a particular column.
- **Sorting Data**
- `df = df.sort_values("Marks", ascending=False)`
- This arranges students from the highest marks to the lowest marks.
- **Ranking Data**
- `df["Rank"] = df["Marks"].rank(`
- `ascending=False`
- `)`
- Sorting helps organize records, while ranking assigns relative positions to the observations.

Data Visualization with Pandas

- After manipulating data, visualization can be used to present the results graphically.
- `import matplotlib.pyplot as plt`
- Different types of plots can be used depending on the purpose of analysis.
- Bar charts are useful for comparing categories.
- Line charts show changes and trends.
- Histograms show the distribution of numerical data.
- Scatter plots show relationships between variables.

Applications of Pandas



- Pandas can be applied to many real-world data analysis tasks.
- In healthcare, Pandas can be used to analyze patient records and medical datasets.
- In education, it can be used to study student performance and attendance.
- In business, it can help analyze sales, customers, and financial records.
- In transportation, it can be used to analyze traffic and vehicle-related data.
- In e-commerce, it can help study product sales and customer behavior.

Conclusion

Pandas provides a powerful and flexible environment for manipulating structured data using Python.

Through Pandas, we can efficiently read datasets, select required information, filter records, handle missing values, sort and rank data, group records, and visualize results.

Therefore, Pandas plays an important role in data preprocessing, exploratory data analysis, and machine learning workflows.

Thank
You!